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EDITORS:
Vladan Djokić
Aleksandra Djordjević
Mladen Pešić
Milica Milojević
Aleksandra Milovanović

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Measuring Built Form Diversity for Housing Tenure Mix: Testing Measures at Seven Wards, London

Zhuoshu He[1], Ye Zhang[2]

[1] Department of Architecture, College of Design and Engineering, National University of Singapore, Singapore, zhuoshuhe@u.nus.edu
[2] Department of Architecture, College of Design and Engineering, National University of Singapore, Singapore, akizy@nus.edu.sg

ABSTRACT

Social and physical diversities have been believed to be connected, but their relations remain ambiguous, and planning scholarship has been fearful to making such connections. Meanwhile, physical diversity has barely been measured with comprehensively considering three aspects of diversity - variety, balance and disparity. Shortcomings in measuring physical diversity may be a cause to disappointing connections between social and physical diversities. To measure physical diversity, attribute selection, way of categorization, and standard of balance are key challenges, and they cannot be settled without setting the purpose of diversity and specific urban contexts. Built-form diversity, a persistent dimension of physical diversity with numerous continuous parameters, is appropriate to explore above challenges. Housing tenure mix is an effective proxy of diverse socioeconomic status, the policy focuses to manage increasing social diversity, and a nexus of built and social environments. This study aims to test different diversity indices for selected built-form parameters at neighbourhoods with various tenure mixes, and explore possible mechanisms connecting built-form diversity and tenure mix. Seven wards with various tenure mixes in London are randomly sampled for this purpose. Results show that building types defined by size, shape and k-means clustering, and plot types defined by size, built-up ratio and k-means clustering may be related to dwelling types, which are correlated to tenure types. Diversities of some parameters of building, plot and street configuration appear relations to tenure mix. They indicate the presence of multiple amenities, fine grain but various characteristics of elements, complete hierarchies of street networks.

Keywords: Social Mixing, Urban Diversity, Urban Form, Diversity Measurement, Classification

INTRODUCTION

Diversity is a prevalent ideal but a controversial topic in urban planning and studies. Typically, urban diversity is defined by its dimensions, primarily social, economic and physical. Social dimension refers to a mix of groups differing in socioeconomic status (SES), ethnicity or culture in a place, such as a neighbourhood (Nathan, 2015; Talen and Lee, 2018). The physical dimension encompasses various physical design, mixed use, and expanded public realm – they are main aspects of the built environment (Fainstein, 2005; Talen, 2006a). On the one hand, each dimension holds its distinct value. For instance, social diversity ensures equal proximity to resources, and nurtures knowledge spillover, aligning with the value of urbanism. Creating diverse built environment remains a common goal in

urban design and planning (ASBEC, 2017). On the other hand, social and physical dimensions of diversity are believed to be linked, but with "*considerable disagreements as to the direction of causality*" (Fainstein, 2005: 4). To assess whether and to what extent physical diversity facilitates social diversity, properly measuring (the degree of) diversity is required.

From this point of view, diversity can also be defined by its numeral pattern. Such a definition encompasses three concepts, namely, variety, balance and disparity (Junge, 1994; Stirling, 2007). It is assumed that a system encompasses plenty of objects, which are partitioned into categories (types) based on attributes. Disparity pertains to the manner and degree by which these objects are distinguished, forming the basis for categorization of objects. Variety (richness) refers to the number of categories. Balance (including evenness and concentration) describes the apportionment of objects across categories. As for measures to these three concepts, variety and balance are often measured by Simpson index, Shannon entropy, and their variations. Sometimes proportions of specific categories are employed as proxies. Disparity for a single, continuous attribute can be measured by standard deviation (SD), coefficient of variation, or their variations. Measures of disparity is often conducted in the format of categorization (typology) – it is a prerequisite to measure variety and balance. Some objects have intrinsic categories (Junge, 1994; Page, 2010; Weitzman, 1992) that are qualitatively different (e.g., ethnicity, land use). While in many cases, categories are defined by continuous attributes (e.g., building size, density), characterized by measurable cross-category distances and fuzzy categorical boundaries (Page, 2010). Clustering algorithms are employed to perform categorization, based on distances of pairwise objects as defined by attributes (Page, 2010; Weitzman, 1992).

However, limited studies have assessed relations between social and physical diversities, and almost no research has considered all the three concepts in measuring physical diversity (He and Zhang, 2023). It remains ambiguous that what elements and characteristics of built environment, as well as to what extent, are associated with various kinds of social diversity. A recent review found measuring diversity was "*underrepresented*" in quantitative research of built form (Fleischmann et al., 2020). Moreover, "*planning scholarship has avoided making these connections … between social goals like diversity and physical design … past experience has been so profoundly negative*" (Talen, 2005: 229). "*Fearful of past mistakes … ideas have rarely been translated into urban design principles, leaving theories on the physical diversity of cities underdeveloped*" (Adams, 2013; see also Talen and Lee, 2018: 50). This prompts a pertinent question: have undeveloped measures of physical diversity been a cause of disappointing findings connecting social and physical diversity?

Measuring diversity presents key challenges, particularly categorization, standard of balance, and attribute selection, all of which require to be considered under specific purpose and urban context. Employing different attributes or methods for categorization can result in variations in balance outcomes for the same set of observations. Achieving a perfectly even apportionment may not reflect reality or the optimum (Salat et al., 2010). Researchers have adopted different standards (e.g., the region's average, experience value) as the standard of balance (Maly, 2000; Roberto, 2016). Selecting and weighting attributes (or the application of multivariate analysis) can significantly influence the outcomes of categorization. These decisions should always be based on the intended purpose of examining physical diversity and the specific urban context. For instance, while categorizing buildings based on size, the choice of cutoff points should vary for different purposes (e.g., relating to building function or capturing numeral distribution only) and in different cities. Similarly, for land use mix, facilitating mixed-income neighbourhoods and promoting active transport require different land use types in different proportions. Some numeral patterns are mathematically feasible, but may not appear in the reality of built environment. In addition, no single index can capture diversity

comprehensively (Livingston et al., 2013), so we always need multiple measures, and compare them when applied to different objects under various conditions (Mcdonald and Dimmick, 2003).

Housing tenure mix is the chosen proxy for social diversity in this study. Increasing social diversity has been an apparent social fact in big cities (Piekut et al., 2012), but it is accompanied by rising intergroup conflicts (Raco and Tasan-Kok, 2019) and exacerbated segregation (Talen and Anselin, 2022). The severe consequence of segregation and the concentration of disadvantaged individuals underscore the importance of mixing, at least to some extent (Tach et al., 2014). Creating diverse communities through deliberate mixing residents in various SES have been adopted as a prominent policy in managing diversity (Talen and Lee, 2018). Within this context, social diversity is often articulated as social mixing, housing tenure mix, or mixed-income housing (Escafré-Dublet and Lelévrier, 2019). SES has been considered a more powerful basis for mixing than ethnicity, as it addresses the fundamental issue of inequality (Galster, 2013; Talen, 2006b). Housing (especially tenure) has not only been the policy focus (Kearns and Mason, 2007), but also an effective proxy of income, the core dimension of SES, in research (Roberts, 2007; Tunstall and Fenton, 2006). Neighbourhood is the primary scale in which mixing policies are implemented. Neighbourhood is also the scale of forming identity, and harbouring physical, social or economic features (Maly, 2000 citing Keller 1968). While creating socially mixed neighbourhoods is not that difficult, their performance (e.g., sustaining diversity, achieving policy goals) is influenced by intertwined social and spatial factors (Tach et al., 2014).

Diversity of built form is our focus within physical diversity in this study. We perceive physical diversity is an instrument, or in Emily Talen's words, a "*condition*" where social diversity "*thrives better*" and is "*sustained longer*" (Talen, 2006a: 242). For instance, a diverse built environment has potential to attract individuals in different SES, may or may not through price, and is suitable for planning and investing in various types of housing projects. Housing is often recognized as a part of the built environment (Papas et al., 2007), making it a possible nexus where social and physical diversity may be linked. Built form focuses on morphological characteristics and spatial patterns of three fundamental elements, plot, building and street (Moudon, 1997). Research of built form aims to identify repeating patterns and comprehend how these elements work together to fulfil human needs (Kropf, 2014). It is the most persistent dimension of the built environment that accommodates other dimensions (Kropf, 2014). Parameters are attributes measuring built form, ranging from size and shape to interface and configuration. Measuring built-form diversity reflects the three above challenges, such as its large number of continuous parameters require exploring categorization and attribute selection.

Besides, building, plot and street are core elements that urban design operates, but built form has yet been the main focus of research about spatial factors for housing tenure mix. For example, existing studies often focus on scales of mixing (e.g., "pepper-potting," street-level, estate-level), façade indistinguishability, separate estate entrance, barriers in public space, etc. (Lawton, 2013; Levin et al., 2014; Roberts, 2007; Tiesdell, 2004). They are undeniably important, but built-form elements deserve more attention.

In this study, we regard facilitating neighbourhood tenure mix as the purpose of measuring built-form diversity. This study aims to 1) test different diversity indices for built-form parameters at neighbourhoods with various tenure mix, and 2) explore possible mechanisms on how built-form diversity may relate to tenure mix. The study could be a starting point in response to above challenges in measuring diversity, including categorization, standard of balance, and attribute selection.

METHODOLOGY

Research design

We conducted our research within the context of London, using a sample of seven wards characterized by varying housing tenure mixes. The electoral ward, a fundamental administrative subdivision of the UK, is used to represent the neighbourhood. Ward-scale demographic data is Ward Atlas 2011 Census in 2014 boundaries dataset by Greater London Authority, retrieved from London Datastore. Household tenure types are recategorized to owner-occupied, social rent, and private rent, corresponding to previous studies (Livingston et al., 2013). We computed the Shannon index for these three tenure types and then randomly selected five wards from five equal intervals. This selection process aimed to ensure comprehensive representation, covering the spectrum from low to high tenure evenness across the city. Additionally, we selected two more wards with similar tenure evenness but different compositions, as illustrated in Figure 1. Seven wards have different profiles – Petts Wood and Knoll (PWK) is a south-eastern suburban ward; Belmont (BL) is a north-western suburban ward; Totteridge (TT) is a north, rural-like ward with brooks and fields; Clock House (CH) is a southern suburban ward; South Bermondsey (SB) locates in the city centre where social rent takes the majority; East Finchley (EF) locates at the inner side of north circular road with a mixed fabric; Queenstown (QT) is a central, complex ward with a big park and many industrial buildings undergoing redevelopment. It is assumed that sample wards are representative, and the current situation of tenure mix is a result shaped by policy, market and other factors jointly.

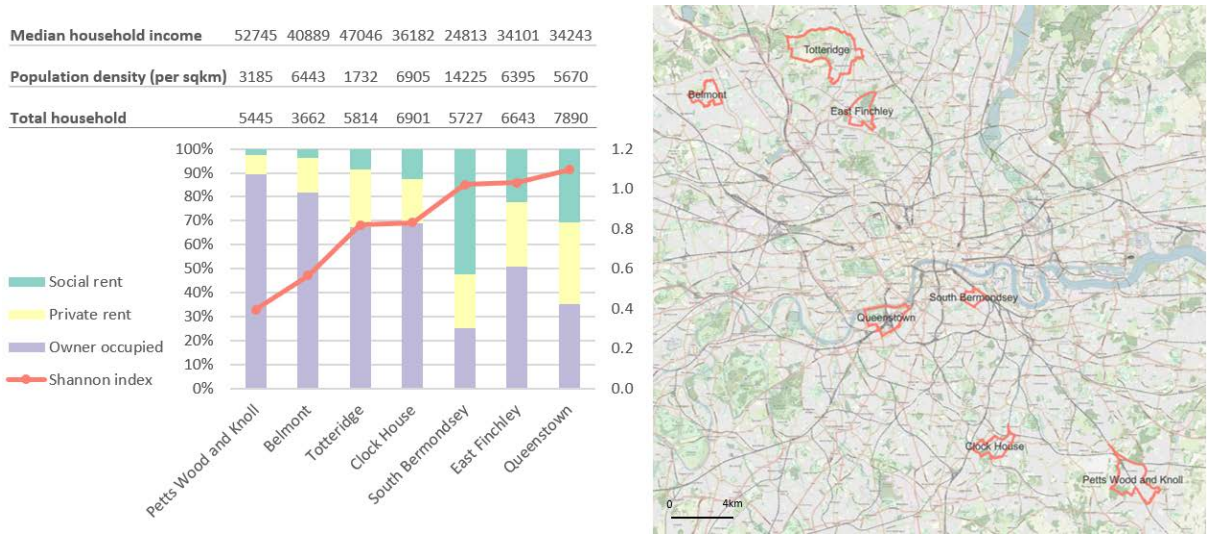


Figure 1. *Sample wards and tenure mix*

Building footprints, street centre lines and plots that intersect with sample wards are observations in this study (Figure 2). Building footprint data is retrieved from Ordnance Survey MasterMap Topography Layer dataset. This is the most precise building footprint data we are able to access, but lack of height, floor number or function. We exclude building footprints smaller than 20 square meters (sqm), as most of them are small, affiliated structures (e.g., garage) that are less functional. Plot (parcel) data is retrieved from INSPIRE Index Polygons spatial data, which has been used by previous studies (Bobkova et al., 2019). Although it does not cover all plots and is messy at some places, it is the most accurate plot data that is open. Our focus is functional plots that may be

relevant to housing, so large parks, fields, water ways, rail corridors are manually excluded. For the same reason, plots smaller than 30 sqm are excluded, because most of them are fragmented polygons that make less sense. Street centre line is retrieved from Ordnance Survey OpenMap Local dataset. Streets within 2 km buffers of sample wards are included for configurational analysis, but only those intersect with sample wards are kept for computing diversity indices. Street centre lines are cleaned and segmentized using QGIS space syntax toolkit, and configurational analysis is conducted in DepthMapX.

For each element, diversity indices of both single and multiple parameters are aggregated at the ward scale, and their results are descriptively compared across sample wards. For single parameter, we first compare numeral patterns across wards using box plots. Second, we conduct categorization and compute Shannon index. We look up not only the Shannon value, but also categorical balance and spatial pattern through stacked bar charts and maps. We employ natural break as the base categorization, but adjust some cutting points if it better aligns with specific purposes (e.g., predict dwelling types). Natural break identifies natural structures of pure numeral patterns, as it categorizes the observations by minimizing the variance within each category and maximizing inter-category variance. For multiple parameters, we employ k-means clustering (Hartigan-Wong algorithm) for each element. K-means clustering has been used for a number of existing studies to explore typology of built form (Berghauser Pont et al., 2019; Schirmer and Axhausen, 2015). Before clustering, parameters are standardized. Shannon index is also computed for clusters, and cross-ward comparison is made. For the convenience of comparison, categorical number is set to be five for all categorizations.



Figure 2. *Built-form elements of the sample wards*

Parameters

Outlines of parameters are shown in Table 1. Three parameters describing size, shape and interface are employed for building footprint. Building size is represented by footprint area, since the only available data is footprint without additional information. Building shape is measured by a parameter named building compactness, which equals to footprint area divided by the area of enclosing circle. It is chosen from a number of shape compactness parameters for plane geometry (Schirmer and Axhausen, 2015). It characterizes irregularity and width-depth ratio at the same time. Interface is measured by what we refer to openness, which equals to the frontage area divided by footprint area. Frontage is defined as the footprint portion that falls inside 10-meter buffer of street centre line. Openness represents how building setback of each street are planned and regulated, and further, the level of street enclosure (Vialard et al., 2021). More built-to-line buildings indicate a compact urban fabric. While (Vialard, 2013) used 20-meter buffer in Atlanta, we test and find that 10-meter better suits London's context, as this distance distinguishes urban and suburban contexts.

For the plot, we use four parameters to describe size, shape, interface and density. Similar to our approach with building footprints, we first consider plot size, which is simply the area. Plot shape compactness equals to plot area divided by the area of enclosing circle. Interface, or plot openness, equals to the length of plot edge directly facing street divided by plot perimeter. The trifold plot parameters are consistent with the framework proposed and utilized by Bobkova et al. (2017). They assumed that a majority of fine-grained plus some varied-size plots established a fertile ground, so as regular-shape plots, as they can easily be split or merged. An extra parameter representing (built-up) density equals to sum of building footprint area within the plot divided by plot area. Sometimes it is also named covered area ratio (Dibble et al., 2019) or ground space index, which characterizes the intensity and compactness (of the urban fabric).

We employ configurational approach for street network analysis, with four parameters, angular choice and angular node count at 400 m and 1500 radii. Metric radii have been shown to be the most effective radius settings (Hillier, 2009). Because we focus on neighbourhood level, 400 m and 1500 m radii (thereafter R400 and R1500) are chosen, in correspondence to local (e.g., walking-distance) and district (e.g., cycling-distance) catchments. Integration measures a street segment's to-movement potential as a destination, while choice measures through-movement potential, or the likelihood to pass through (Hillier, 2009). Human movement follows least angle change paths (Hillier, 2009), thus angular choice helps identify the main route network and hierarchy (Van Nes and Yamu, 2021). In our sample, angular choice is highly correlated ($p < 0.001$) with metric choice (R1500: $r^2 = 0.92$; R400: $r^2 = 0.96$). Node count is "*the number of segments encountered on the route from the current segment to all others*" (Al Sayed et al., 2016: 47). On a small radius, angular integration approximate node count (Al Sayed et al. 2016). In our sample, node count is highly correlated ($p < 0.001$) with angular integration (R1500: $r^2 = 0.88$; R400: $r^2 = 0.80$), angular total depth (R1500: $r^2 = 0.88$; R400: $r^2 = 0.86$), metric total depth (R1500: $r^2 = 0.98$; R400: $r^2 = 0.97$), and metric total length (R1500: $r^2 = 0.97$; R400: $r^2 = 0.93$). Therefore, angular choice and angular node count may be the most representative. We do not use normalized parameters (e.g., NACH, NAIN), but absolute ones. This study aims to compare aggregated indices at ward level, rather than looking at each street's relative hierarchy in different urban contexts. Normalized parameters "*enable the comparison between systems of different sizes*" (Al Sayed et al., 2016: 45). "*When all the maps are georeferenced with the same type of units, there is in principle no need to normalize the values*" (Van Nes and Yamu, 2021: 79). We tested normalized parameters, but find cross-ward differences are diluted.

Table 1. Parameters describing characteristics of built-form elements

Parameter	N	Range	Mean	Median	Standard Deviation
Building	29580				
Footprint Size		20 - 18786.15	88.61	59.08	244.79
Compactness		0.02 - 1	0.50	0.53	0.11
Openness		0 - 1	0.07	0	0.18
Plot	24574				
Size		30 - 159954.11	492.60	276.16	2143.28
Compactness		0.01 - 0.81	0.29	0.27	0.10
Openness		0 - 1	0.14	0.09	0.14
Built-up Density		0 - 1	0.28	0.24	0.16
Street	3135				
Angular choice R = 400		0 - 1246	107.2	52	145.29
Angular Choice R = 1500		0 - 64036	5480.4	1388	9073.84
Node Count R = 400		3 - 119	39.07	37	18.89
Node Count R = 1500		4 - 1189	509.3	489	213.83

RESULTS AND DISCUSSIONS

Building

To measure building and plot diversity for tenure mix, we assume that types defined by building and plot parameters may potentially predict dwelling types, and dwelling types predict tenure types. London's ward-scale data (n = 630) shows that owner-occupied has significantly positive correlation ($p < 0.001$) with detached ($r^2 = 0.6$), semi-detached ($r^2 = 0.82$); and significantly negative correlation with flat ($r^2 = -0.84$, $p < 0.001$). Both private rent and social rent have significantly positive correlation ($p < 0.001$) with flat ($r^2 = 0.69$ and $r^2 = 0.56$ respectively). In this way, the result of each parameter (or the element's k-mean clustering) is presented twofold. First, we explore categorization and its possible relations with dwelling types. Second, we look up descriptive statistics, diversity indices, and type compositions, exploring their possible connections with tenure mix from observed patterns.

Building footprint size may predict dwelling types to some extent, and diversity of footprint size has somewhat positive association with tenure mix. First, footprint size may distinguish dwelling types. Natural break cutting points for footprint size are 197, 632, 1665, 4586 sqm, and for logged footprint size are 38, 63, 117, 361 sqm, both of which make less sense in distinction across dwelling types.

Through observation, we tentatively set cutting points at 50, 150, 500, 2000 sqm. In this way, most landed houses (type 1 and 2) and apartments (type 3 and 4) are largely distinguished. There are some row houses with simple and rectangular footprint shape, which look like being developed collectively (type 1). Observed from the map, besides apartments, collective row houses may be associated with social rent. Terraced houses and the majority of detached and semi-detached houses have similar sizes (type 2). Some other semi-detached houses, mainly at PWK, BM and CH, have small footprints (type 1). Some detached houses at TT have larger footprint (type 3). Type 3 also includes some amenities (e.g., shophouse, church). Schools and industrial buildings (e.g., warehouses) have large footprints (type 4, 5).

Second, footprint size SD and type evenness have somewhat positive association with tenure mix (Figure 3). Type 1 and 2 are always the majority, thus more large-footprint buildings lead to increased evenness. From map observation we can interpret that more large-footprint buildings indicate more amenities and apartments, which are signs of functional and dwelling mixes.



Figure 3. *Building footprint size*

Building compactness types are less predictable to dwelling types, but diversity of building compactness may have positive relations to tenure mix, while mean may have negative relations. First, building compactness types are less predictable to dwelling types, hence tenure types, mainly due to dwelling types associated with different tenure types overlap in building compactness types. For example, collective row houses, semi-detached houses and detached houses have close-to-square shapes (mixed in type 4 and 5). Many apartments in modern era have elongated shapes (type 1 and 2), but some apartments at the compact areas are smaller (type 3 and 4). Terrace houses often fall in type 2. Type 3 mixes above dwelling types. Non-residential functions do not show clear pattern. They may be distinguished by other shape parameters, like skeleton, corner or bounding box, which capture irregularity or complexity more accurately. It seems type 1 and 2 are positively related to tenure mix, as they mainly relate to apartments. An apartment building, with many units, has more effect on housing balance than a landed house.

Second, building compactness SD and evenness of types shows positive association with tenure mix (Figure 4). If we combine type 4 and 5 (which relate to same dwelling types), this association becomes more significant, showing the importance of categorization in measuring balance. In addition, mean of building compactness has negative association with tenure mix. A larger number of “thinner” buildings indicate a denser, compacter urban fabric, which may be suitable for tenure mix.

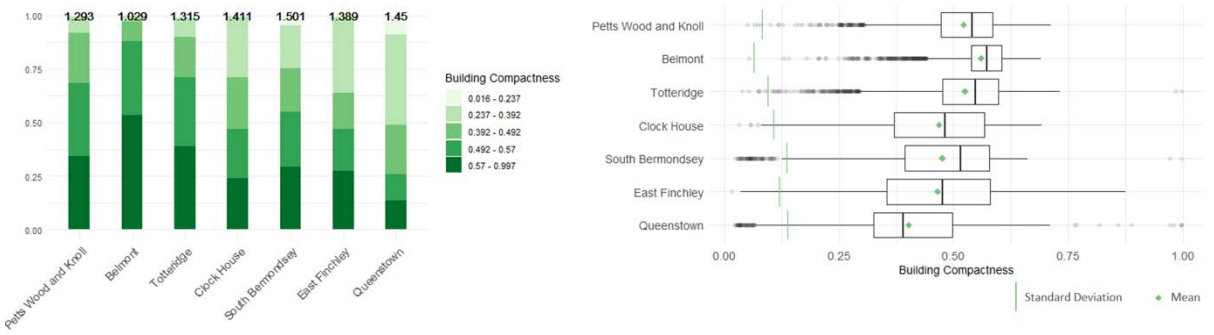


Figure 4. *Building shape compactness*

Diversity and mean of building openness appear to be positively associated with tenure mix. First, building openness may not directly relate to dwelling or tenure types. The first cutting point determined by natural break is 0.72. However, to distinguish buildings with more than 10-meter setback from those with roughly 10-meter setback, but we adjust it to 0.01, and merge last two types as they constitute only small portions.

Second, proportion of building with more than 10-meter setback is negatively associated with tenure mix, and mean openness is positively related to tenure mix (Figure 5). Evenness of openness types is also positively related to tenure mix. Openness is related to not only frontage, but also building depth. Therefore, presenting more buildings with less than 10 meter and varied setbacks, and finer grains (e.g., smaller building depth) may facilitate tenure mix. Similarly, having many streets characterized by overall high yet various enclosure (height-width ratio of street sections) may be also a facilitator. In addition, the current setback is calculated from street centre line without distinguishing street hierarchies, due to the data restriction. If curb line data is available, the performance of building openness parameter can be improved.

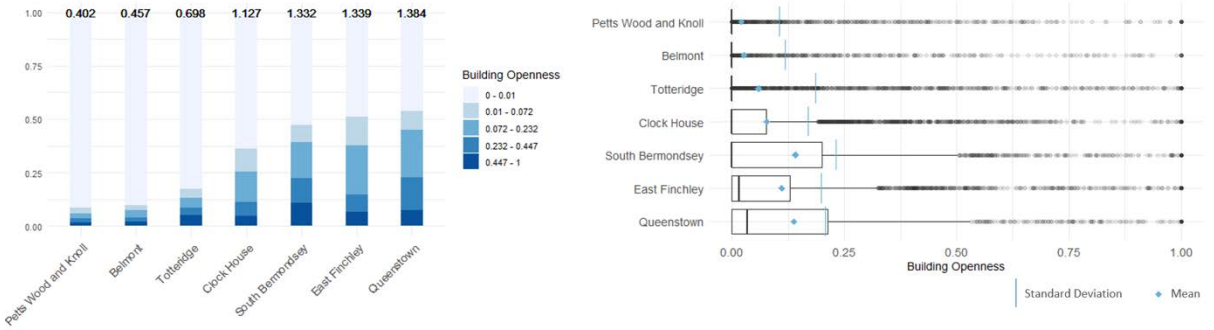


Figure 5*. Building openness*

Evenness of k-means clusters appears to be positively associated with tenure mix (Figure 6). K-means clusters do not better relate to dwelling types than footprint size. Some loose connections could be drawn, such as type 4 with terraced houses, type 1 with detached, semi-detached and collective row houses, and type 5 with large, non-landed-house buildings. The continuous nature of the parameters results in significant overlap between clusters. In the map, it can be difficult to specify characteristic differences between clusters. Nevertheless, the result still suggest higher tenure-mix neighbourhoods are likely to have diverse building form.

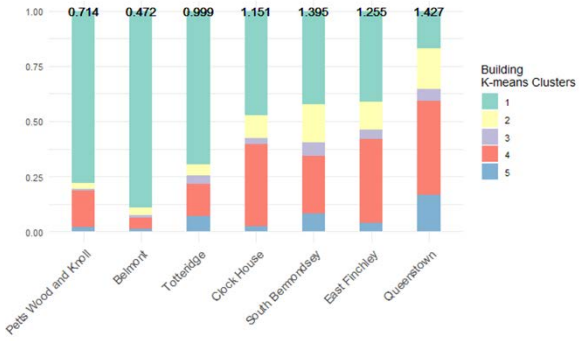


Figure 6. *Building k-means clusters*

Plot

Plot size diversity appears somewhat positive relations with tenure mix. First, link between plot size and dwelling type is less clear compared to building size. For the same dwelling type, the nearer it locates to the city centre, the smaller its plot size is - this pattern is more significant for plot than building footprint. In other words, different dwelling types have large overlaps in plot size. Natural breaks cutting points are approximately 147, 297, 596, 2094. Through map observation and testing, we set cutting points at 200, 500, 1000, 4000. In this way, most collective row houses and terraced houses fall in type 1 and 2; most semi-detached houses fall in type 2 and 3; most detached houses fall in type 3, 4 and 5 (type 5 are rural ones at TT); and apartments are often type 4 and 5.

Second, plot size SD and high proportions of small (type 1) and large (type 5) plots may be positively related to tenure mix, but evenness of types does not appear a significant relation (Figure 7). If we take ward location into account, for example, comparing two pairs of wards in similar location (CH and EF; SB and QT), wards with higher tenure evenness have higher plot size SD. It is interpreted as finer grain (more terraced, less semi-detached or detached houses, and smaller plots), along with presence of more apartments, may facilitate tenure mix.

For both building size and plot size, location impacts their relations with tenure mix. Proximity to city centre drives median lower, but mean and SD higher. Median represents the most common plot, usually landed house. Suburban or rural neighbourhoods have more detached and semi-detached houses, as well as larger grain for the same dwelling type. In contrast, central neighbourhoods' landed houses have finer grain (smaller median), but they have higher density hence more larger buildings and plots (higher mean and SD).



Figure 7. *Plot size*

Plot (built-up) density types may predict dwelling types, and plot density and its diversity show positive relations with tenure mix. First, plot density types are related to dwelling types. We adjust natural breaks cutting points (0.161, 0.289, 0.453, 0.686) to 0.1, 0.3, 0.4, and 0.5. Apartments mainly fall into type 2, with a few in type 3 (at central wards like SB). Collective row houses mainly fall into type 2, 3. Terraced houses mainly fall into type 3, 4, and a small part in type 5 (at QT and EF). Semi-detached and detached houses mainly fall in type 2, but some detached houses at low-density areas (e.g., TT and PWK) fall in type 1.

Second, mean, median, SD and type evenness of plot density are positively related to tenure mix (Figure 8). The relation is blurred, because again, dwelling types relating to different tenures overlap in same density types (e.g., type 2 mixes apartments, semi-detached and detached plots). For wards in similar locations but different tenure mixes, more mixed ones are denser (QT is denser than SB, and EF is denser than CH).

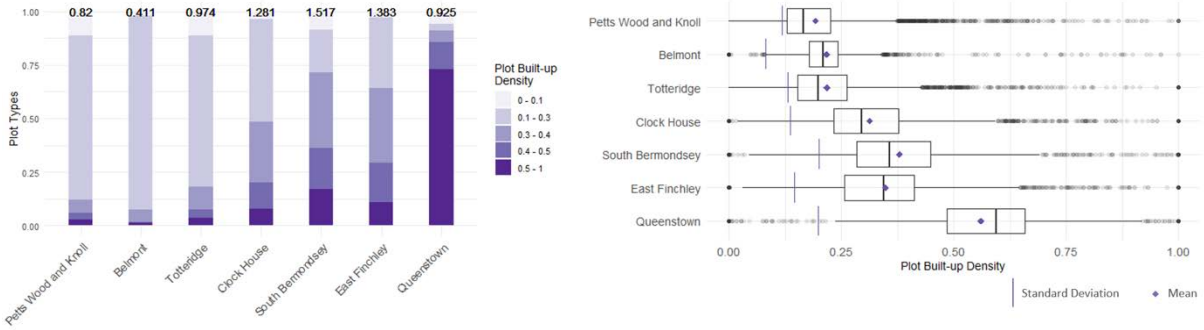


Figure 8. *Plot built-up density*

Diversity of both compactness and openness do not show clear relation to tenure mix. First, plot compactness types do not show a clear pattern relating to tenure type. As for openness, most landed-house plots fall in type 2, except for those at the street intersection, which fall in type 4. Most apartment plots (one estate is often one plot) fall in type 3-5.

Second, diversity and statistic of compactness and openness do not appear clear relation to tenure mix, even contradict (Figure 9 - 10). Taking median as an example. The median value should appear at a typical plot (rectangle, short edge faces street), both compactness and openness values are negatively related to depth-width ratio. However, medians of compactness and openness appear contradicted patterns. Proximity to the city centre means width and depth are both smaller, so it is uncertain how this two-way effect acts. We find depth-width ratio of the typical plot may have some repeated patterns, but they have not been revealed.

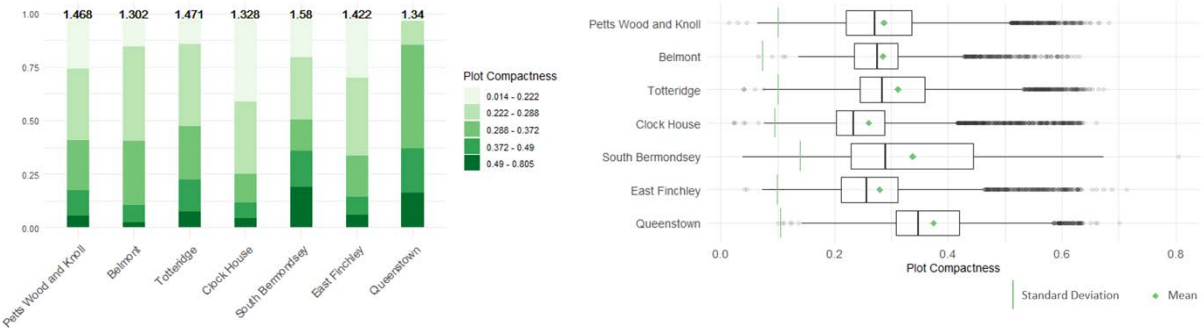


Figure 9. *Plot compactness*

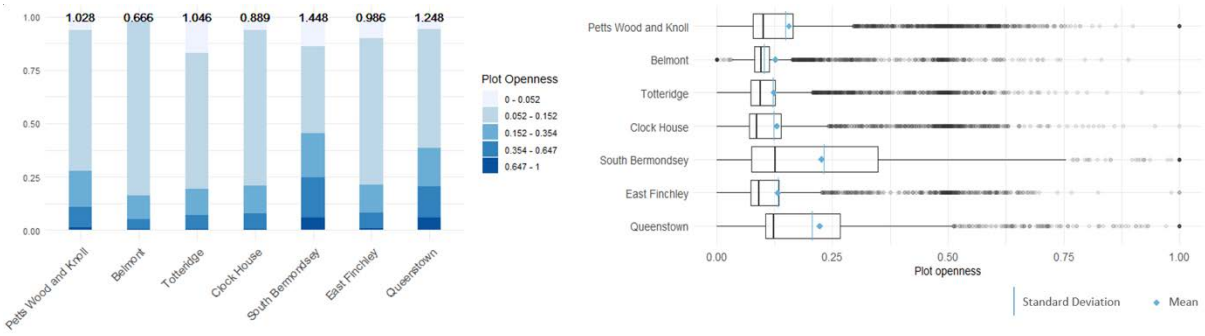


Figure 10. *Plot openness*

Plot k-mean clusters are somewhat related to dwelling types, and the proportions of certain types appear to be positively related to tenure mix. First, plot k-mean clusters may relate to dwelling types, but relations may not be stronger than single parameters like size or density. Apartments fall in type 1 (lower openness) and 2 (higher openness). Terraced and collective row houses fall in type 4 (larger depth) and 5 (smaller depth, mainly at QT). Semi-detached houses mainly fall in type 3, while detached houses often fall in type 2 (more rural, larger, and more irregular plots) and 3.

Second, evenness of k-means clusters does not show a clear relation to tenure mix, as wards are dominated by different types, but proportion combining type 4 and type 5 may be positively related to tenure mix (Figure 11). Dominance of type 4 and 5 (mainly terraced and collective row houses) is a sign of more compact fabric, in contrast to type 3 (semi-detached and detached houses). Similar to building, plot k-means clusters sometimes have difficulties in interpreting characteristic differences between clusters from the map, which due to large cross-clusters overlaps. Combination of parameters to define plot types can be improved. In addition, incomplete plot data and lack of plot-scale land use information are main restrictions.

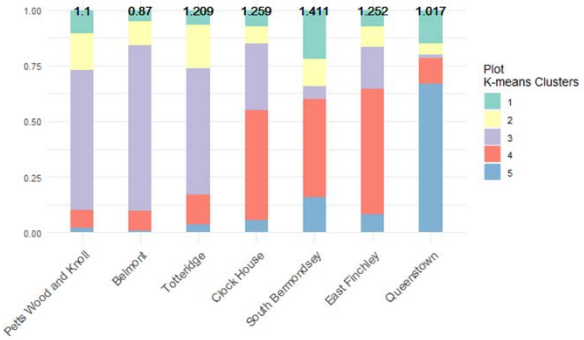


Figure 11. *Plot k-means clusters*

Street

Diversities of angular choice (ACH) on both radii may be positively related to tenure mix, and they identify different levels' major streets to some extent. First, ACH on R1500 and R400 are correlated ($r^2 = 0.71$, $p < 0.001$), but have some different spatial patterns. ACH R1500 identifies hierarchies and district-or-above-level major streets clearer, while high ACH R400 identifies some local-level major street. ACH on both radii may not spatially relate to different dwelling types. We split ACH = 0 (not

falling on any shortest path) from the first type set by natural breaks, and merge the last two types as they constitute only small portions.

Second, for both radii, mean, SD, and type evenness are positively related to tenure mix, but the relation may drop at highest tenure-mix wards (Figure 12 - 13). This suggests that mixed-tenure wards are likely to locate around convenient, major, and high through-movement potential streets at both local and district levels. Meanwhile, mixed-tenure wards have streets in various ACH ranges, indicating complete hierarchies. Of course, location mediates these relations. For instance, the river and the park get in the way of QT's connections to the surrounding area, especially for R1500 distance, which may be a cause to the drop of mean and diversity.

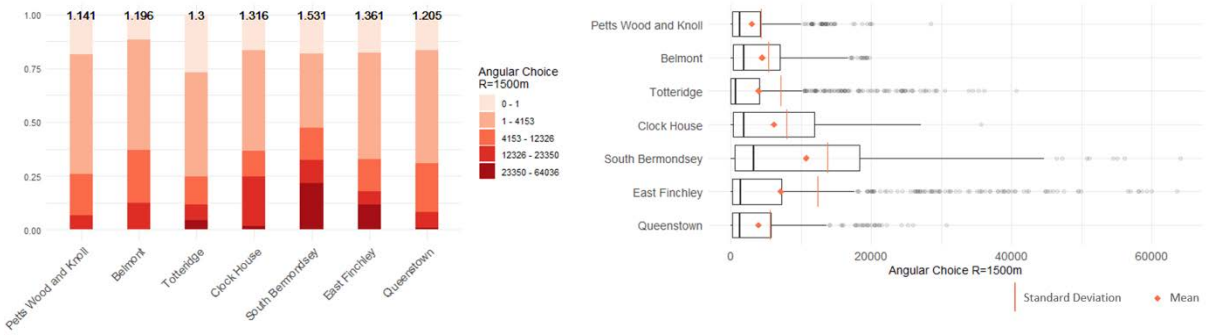


Figure 12. *Segment angular choice R = 1500 m*

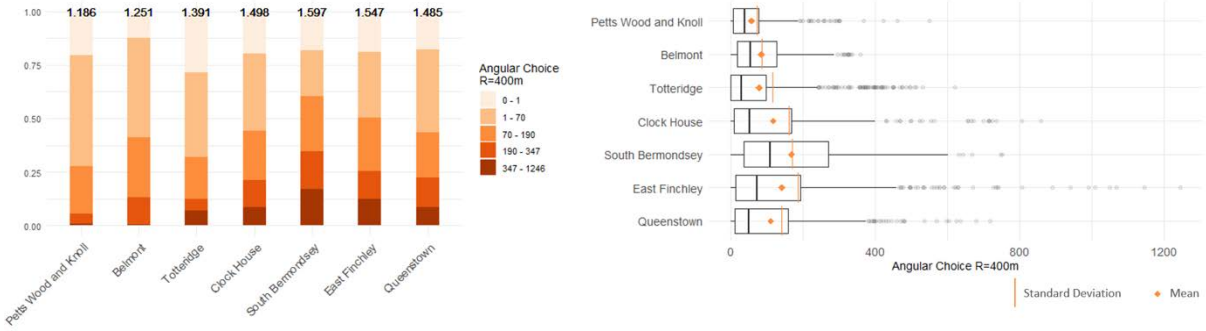


Figure 13. *Segment angular choice R = 400 m*

Node count (NC) diversity on R400 and R1500 are positively related to tenure mix, but R400 is more significant. First, high NC R400 identifies main roads entering compact estates (or small residential areas) and parks, as well as roads accessing amenities. These spaces happen to be located at high to-potential streets. Although NC R1500 is correlated with NC R400 ($r^2 = 0.71$; $p < 0.001$), R1500 presents less clear spatial patterns, and more sensitive to location and context. For example, SB locates at a well-connected area, so its NC on R1500 stands out from all wards.

Second, NC R400's SD and type evenness, as well as NC R1500's SD are positively related to tenure mix (Figure 14 - 15). Having streets with various to-movement potential or accessibility at local level may create spatial conditions for tenure mix. High NC streets also show spatial potentials to be social seams, which are crucial for successful tenure mix (Sautkina et al., 2012). For both ACH and NC, R400 is more significant than R1500 in terms of relating tenure mix.

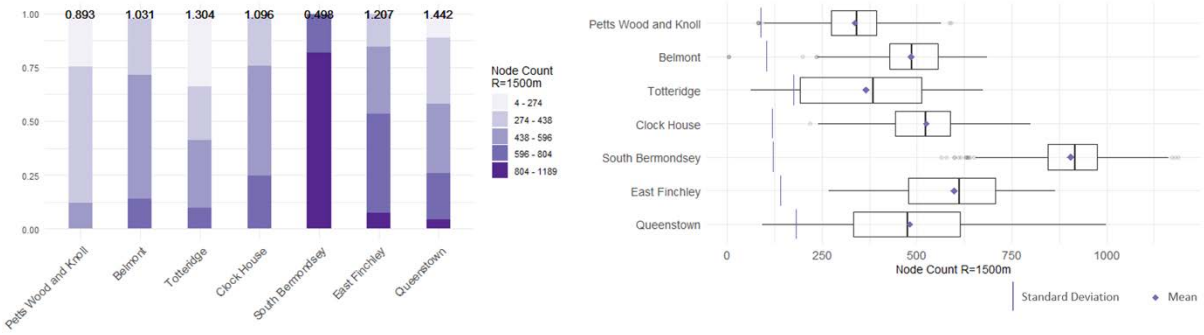


Figure 14. *Segment node count R = 1500 m*

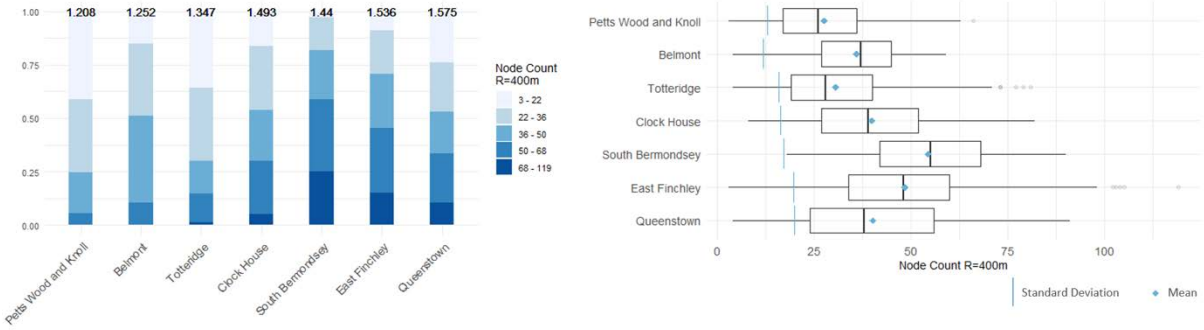


Figure 15*. Segment node count R= 400 m*

Evenness of k-means clusters does not show significant relations with tenure mix. First, street k-means clusters do not offer better to street hierarchies than single parameter. Type 1 is often well-connected major street segments at local or district levels. Type 2 is less well-connected than type 1, including some segments of major streets and streets on a lower hierarchy. Type 3 is mainly branch streets in small residential areas, with both edges connected to the above two types. Type 4 and 5 are often dead ends.

Second, as shown in Figure 16, higher tenure-mix wards may have more well-connected streets (e.g., type 1), although impacted by location (e.g., park and river surrounding QT). Besides, this clustering only applies a few configurational parameters, from a huge number of them on different radii. Limitations in testing relations across parameters and selecting appropriate ones (e.g., employing multivariate analysis) could contribute to the somewhat less-positive results obtained.

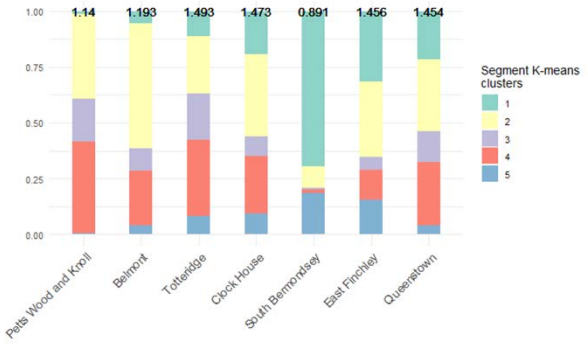


Figure 16. *Segment k-means clusters*

CONCLUSIONS

In this study, we test diversity indices for built-form parameters at neighbourhoods with various tenure mixes, as a preliminary exploration in challenges of measuring physical diversity, including categorization, standard of balance, and attribute selection. Based on the assumption that dwelling types are associated with tenure types, building types and plot types defined by parameters may claim a purpose of predicting dwelling types. This is parallel to another way of categorization, with the purpose of capturing the numeral patterns (e.g., natural break, clustering). This study finds that building size, building shape, plot size and plot built-up density may predict dwelling types to different extents (Table 2). For this purpose, K-means clusters of building and plot do not perform better than single parameter. There should be more research on comparing different ways of categorizing built-form elements, to what extent they predict dwelling or functional types, and their effects on measuring balance and diversity. We need smarter combining multiple parameters in categorization values in future research. Additionally, higher accuracy of data, like building footprint with floor number or accurate tenure location, will help exploration of building-dwelling-tenure relations.

Table 2. *Relations between types defined by parameters and dwelling types*

Type defined by parameter	Amenities	Apartments	Collective-developed row houses	Terraced houses	Semi-detached houses	Detached houses
Building size type	3, 4, 5	3, 4	1	2	1, 2	2, 3
Building compactness type	-	1, 2, 3, 4	4, 5	2	4, 5	4, 5
Building k-means cluster	5	2, 5	1, 2	2, 4	1	1
Plot size type	2, 3, 4, 5	4, 5	1, 2	1, 2	2, 3	3, 4, 5
Plot density type	-	2	2, 3	3, 4, 5	2	1, 2
Plot k-means cluster	-	1	3, 4	4	3, 4	2, 3

Besides categorization related to dwelling types and then tenure types, diversity and statistics of built-form parameters indicate the neighbourhood's fabric characteristics, which can be facilitator for tenure mix. We find some descriptive statistics may be related to tenure mix, for example, mean or median of building openness, plot built-up density, and street angular choice. They often show overall compact fabric, with many built-to-line streets, and well-connected networks. Moreover, diversity of building footprint size, building compactness, building openness, plot size, plot built-up density, street angular choice, and street node count may be related to tenure mix. They indicate, especially for residential-dominant neighbourhoods, the presence of multiple amenities and ranges of various built-form characteristics (e.g., hierarchies of public spaces and streets) may facilitate tenure mix.

Future research will test relations between built-form diversity and tenure mix through statistical models, with taking into account other factors (e.g., location, neighbourhood SES). The basis is to find out effective diversity indices for tenure mix, with addressing categorization, standard of balance, and attribute (parameter) selection or combination. Based on the small samples of this study, SD and type evenness often have similar pattern in relating to tenure mix, and sometimes percentage of certain type can be more effective. One thing should be noted is that these relations may be non-linear (Dovey and Pafka, 2017; Kearns and Mason, 2007) or not direct but through mediators like social network (Talen and Lee, 2018). Therefore, we also need detailed investigation and observation of patterns, as well as review of policy and planning regulations. As such we are able to set clearer condition for measuring diversity, analysing possible mechanism and reflecting to policy in-turn. "*Qualitative studies are able to identify and explore channels that quantitative research then systematically tests*" (Nathan, 2015: 7).

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